

# Closed-Form Richardson-Number Stability Functions from Monin–Obukhov Similarity Theory: Re-Derivation, Generalization, and Implementation Notes

*A Technical Working Reference for Boundary-Layer Modeling*

**Department of Atmospheric & Earth Science**

University of Alabama in Huntsville (UAH)

*Revising and Extending England & McNider (1995)*

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## Abstract

This document provides a mathematically rigorous, fully self-consistent reference for transforming Monin–Obukhov Similarity Theory (MOST) flux-profile functions  $\phi_m(\zeta)$  and  $\phi_h(\zeta)$  into explicit, closed-form gradient Richardson number ( $Ri$ ) stability functions  $f_m(Ri)$  and  $f_h(Ri)$ . We document and correct a historical transcription error in the  $\zeta \leftrightarrow Ri$  mapping relation found in early literature, highlight the general master equation governing the stable boundary layer (SBL) for arbitrary heat profile exponents  $\alpha_h$ , detail the step-by-step derivation of the unstable branch cubic equation, derive near-neutral series expansions via series reversion, and provide an explicit integration of surface roughness  $z_0$  for discrete boundary-layer models.

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## 1 Notation & Nomenclature

To reduce cognitive load and prevent ambiguity across local point-gradient theory, bulk atmospheric grid layers, and numerical implementations, the notation defined in Table 1 is maintained throughout this memorandum.

## 2 Theoretical Foundations & Coupling Identities

In first-order eddy-diffusivity closures ( $K$ -theory), turbulent exchange coefficients for momentum ( $K_m$ ) and heat ( $K_h$ ) are parameterized using mixing-length theory driven by mean vertical shear  $S = |\partial u / \partial z|$ :

$$K_m = l_m^2 S f_m(Ri), \quad K_h = l_h^2 S f_h(Ri) \tag{1}$$

Table 1: Summary of Mathematical Symbols and Definitions

| Symbol           | Description                                | Definition / Relation                                                                                                               |
|------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $Ri_g$           | Gradient Richardson Number                 | $\frac{g}{\theta_0} \frac{\partial \theta / \partial z}{(\partial u / \partial z)^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)}$ |
| $Ri_b$           | Bulk Layer Richardson Number               | $\frac{g}{\theta_0} \frac{\Delta \theta \Delta z}{(\Delta u)^2}$                                                                    |
| $Ri_c$           | Critical Richardson Number                 | $\lim_{\zeta \rightarrow \infty} Ri_g(\zeta) = \frac{1}{\beta_m}$ (Limiting value for turbulence)                                   |
| $Ri$             | Generic Local Richardson Number            | Used in general algebraic derivations ( $Ri \equiv Ri_g$ )                                                                          |
| $\zeta$          | Dimensionless Monin–Obukhov Stability      | $z/L$                                                                                                                               |
| $L$              | Monin–Obukhov Length Scale                 | $\frac{u_*^2 \theta_0}{\kappa g \theta_*}$                                                                                          |
| $f_m, f_h$       | $Ri$ -Based Stability Correction Functions | $K_m = l_m^2 S f_m(Ri)$ , $K_h = l_h^2 S f_h(Ri)$                                                                                   |
| $\phi_m, \phi_h$ | MOST Non-Dimensional Profile Functions     | $\frac{\kappa z}{u_*} \frac{\partial u}{\partial z}$ , $\frac{\kappa z}{\theta_*} \frac{\partial \theta}{\partial z}$               |

where  $l_m = l_h = \kappa z$  near the surface. Conversely, Monin–Obukhov Similarity Theory (MOST; Businger et al. 1971; Dyer 1974) defines diffusivities as:

$$K_m = \frac{\kappa z u_*}{\phi_m(\zeta)}, \quad K_h = \frac{\kappa z u_*}{\phi_h(\zeta)} \quad (2)$$

Equating  $K$ -theory with MOST using the surface-layer profile relation  $S = \frac{u_*}{\kappa z} \phi_m(\zeta)$  yields the fundamental coupling definitions:

$$f_m(Ri) = \frac{1}{\phi_m^2(\zeta)} \quad (3)$$

$$f_h(Ri) = \frac{1}{\phi_m(\zeta) \phi_h(\zeta)} = \frac{f_m^{1/2}(Ri)}{\phi_h(\zeta)} \quad (4)$$

## 2.1 Correction of the Historical $\zeta \leftrightarrow Ri$ Transformation

The gradient Richardson number  $Ri_g$  is defined via MOST profiles as:

$$Ri = \frac{g}{\theta_0} \frac{\frac{\partial \theta}{\partial z}}{\left(\frac{\partial u}{\partial z}\right)^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} \quad (5)$$

Substituting (3) and (4) into (5) yields the exact relation connecting  $Ri$  and  $\zeta$ :

$$\phi_m(\zeta) = f_m^{-1/2}(Ri), \quad \phi_h(\zeta) = \frac{f_m^{1/2}(Ri)}{f_h(Ri)} \quad (6)$$

Inserting these back into (5):

$$Ri = \zeta \frac{f_m^{1/2}/f_h}{f_m^{-1}} = \zeta \frac{f_m^{3/2}(Ri)}{f_h(Ri)} \implies \boxed{\zeta = Ri \frac{f_h(Ri)}{f_m^{3/2}(Ri)}} \quad (7)$$

### 3 Stable Boundary Layer Formulation ( $\zeta \geq 0$ )

We generalize the stable non-dimensional profiles by retaining a linear momentum function ( $\alpha_m = 1$ ) while allowing a general exponent  $\alpha_h$  for heat:

$$\phi_m(\zeta) = 1 + \beta_m \zeta, \quad \phi_h(\zeta) = (1 + \beta_h \zeta)^{\alpha_h} \quad (\zeta \geq 0) \quad (8)$$

Defining  $x \equiv f_m^{1/2}(Ri) = \phi_m^{-1}(\zeta)$ , we invert  $\phi_m(\zeta)$  to obtain:

$$\zeta = \frac{1-x}{\beta_m x} \quad (9)$$

#### 3.1 Central Master Equation in $x \equiv f_m^{1/2}$

Substituting (9) into (7) gives the explicit expression for  $f_h(Ri)$ :

$$f_h(Ri) = \frac{x^2(1-x)}{\beta_m Ri} \quad (10)$$

Equating  $\phi_h = x/f_h = \frac{\beta_m Ri}{x(1-x)}$  with  $\phi_h(\zeta) = \left(1 + \beta_h \frac{1-x}{\beta_m x}\right)^{\alpha_h}$  produces the **central master equation** governing all stable FIRST-order closures:

$$\boxed{\left[ \frac{Ri \beta_m}{x(1-x)} \right]^{1/\alpha_h} = \frac{\beta_m x + \beta_h(1-x)}{\beta_m x}} \quad (11)$$

#### 3.2 Case 1: Standard Linear Heat Profile ( $\alpha_h = 1$ )

Setting  $\alpha_h = 1$  reduces (11) directly to a quadratic polynomial in  $x$ :

$$Ri(\beta_h - \beta_m)x^2 - (1 + Ri\beta_h)x + 1 = 0 \quad (12)$$

Taking the physically meaningful root ( $x \rightarrow 1$  as  $Ri \rightarrow 0$ ) yields the exact closed-form momentum stability function:

$$f_m(Ri) = x^2 = \left[ \frac{(1 + Ri\beta_h) - \sqrt{(1 + Ri\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)} \right]^2 \quad (13)$$

The corresponding heat function  $f_h(Ri)$  is evaluated directly via (10).

### 3.3 Case 2: Businger–Dyer Quadratic Heat Profile ( $\alpha_h = 2$ )

Setting  $\alpha_h = 2$  and squaring both sides of (11) clears fractional exponents, yielding a cubic polynomial  $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$ :

$$\begin{aligned} a_3 &= -(\beta_m - \beta_h)^2 \\ a_2 &= (\beta_m - \beta_h)^2 - 2\beta_h(\beta_m - \beta_h) \\ a_1 &= 2\beta_h(\beta_m - \beta_h) - \beta_h^2 - Ri\beta_m^3 \\ a_0 &= \beta_h^2 \end{aligned} \tag{14}$$

### 3.4 Equal-Parameter Limit ( $\beta_m = \beta_h = \beta$ )

When  $\beta_m = \beta_h = \beta$ ,  $a_3 = 0$  and  $a_2 = 0$ . The quadratic (12) degenerates to a simple linear equation in  $x$ :

$$(1 + \beta Ri)x = 1 \implies x = \frac{1}{1 + \beta Ri} \tag{15}$$

which yields:

$$\boxed{f_m(Ri) = (1 + \beta Ri)^{-2}, \quad f_h(Ri) = (1 + \beta Ri)^{-3}} \tag{16}$$

*Distinction Note:* This result demonstrates that quadratic closures like  $(1 - \beta Ri)^2$  stem from distinct physical assumption sets ( $Pr_t = 1$  toy models) rather than standard linear MOST profile inversions.

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## 4 Unstable Boundary Layer Formulation ( $\zeta < 0$ )

For unstable stratification ( $\zeta < 0$ ), standard Businger–Dyer profile functions take the generalized exponential form:

$$\phi_m(\zeta) = (1 - \beta_m \zeta)^{-\alpha_m}, \quad \phi_h(\zeta) = (1 - \beta_h \zeta)^{-\alpha_h} \tag{17}$$

For classical coefficients ( $\alpha_m = 1/4$ ,  $\alpha_h = 1/2$ ), the stability functions map in terms of  $\zeta$  as:

$$f_m(\zeta) = (1 - \beta_m \zeta)^{1/2} \tag{18}$$

$$f_h(\zeta) = (1 - \beta_h \zeta)^{1/2} (1 - \beta_m \zeta)^{1/4} \tag{19}$$

### 4.1 Master Cubic Equation in $x \equiv f_m^2$

Expressing  $f_h$  directly in terms of  $f_m$ :

$$f_h(f_m) = f_m^{1/2} \left[ 1 - \frac{\beta_h}{\beta_m} (1 - f_m^2) \right]^{1/2} \tag{20}$$

Substituting (20) into the corrected transformation identity  $\zeta^2 = Ri^2 \frac{f_h^2}{f_m^3}$  and defining  $x \equiv f_m^2(Ri)$  yields the exact master cubic equation for unstable conditions (see Appendix A for full intermediate steps):

$$x^3 - 2x^2 + \left(1 - Ri^2 \beta_m \beta_h\right) x - Ri^2 \beta_m (\beta_m - \beta_h) = 0 \quad (21)$$

#### 4.2 Analytical Solution for Equal Coefficients ( $\beta_m = \beta_h = 16$ )

When  $\beta_m = \beta_h = 16$ , the constant term in (21) vanishes. Factoring out  $x \neq 0$ :

$$x^2 - 2x + (1 - 256Ri^2) = 0 \implies (1 - x)^2 = 256Ri^2 \implies x = 1 - 16Ri \quad (22)$$

Recalling  $x \equiv f_m^2$ , we recover the analytical expression:

$$f_m(Ri) = (1 - 16Ri)^{1/2} \quad (23)$$

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## 5 Series Reversions & Pedagogical Models

### 5.1 Near-Neutral Series Expansions via Series Reversion

When exact inversion is mathematically intractable, Taylor expanding  $\phi_m(\zeta) = 1 + \beta_m \zeta + \gamma_m \zeta^2$  and  $\phi_h(\zeta) = 1 + \beta_h \zeta + \gamma_h \zeta^2$  around  $\zeta = 0$  provides a forward series for  $Ri(\zeta)$ :

$$Ri_g(\zeta) = \zeta + (\beta_h - 2\beta_m)\zeta^2 + (3\beta_m^2 - 2\beta_m\beta_h - 2\gamma_m + \gamma_h)\zeta^3 + O(\zeta^4) \quad (24)$$

Reverting this series yields the inverted expression for  $\zeta(Ri_g)$ :

$$\zeta(Ri_g) = Ri_g + (2\beta_m - \beta_h)Ri_g^2 + (5\beta_m^2 - 6\beta_m\beta_h + 2\beta_h^2)Ri_g^3 + O(Ri_g^4) \quad (25)$$

Substituting  $\zeta(Ri_g)$  into  $f_m = \phi_m^{-2}$  and  $f_h = (\phi_m \phi_h)^{-1}$  yields the low-order  $Ri$ -series forms:

$$f_m(Ri_g) = 1 - 2\beta_m Ri_g - \beta_m(\beta_m - 2\beta_h)Ri_g^2 + O(Ri_g^3) \quad (26)$$

$$f_h(Ri_g) = 1 - (\beta_m + \beta_h)Ri_g - (\beta_m^2 - 2\beta_h^2 + \gamma_m + \gamma_h)Ri_g^2 + O(Ri_g^3) \quad (27)$$

### 5.2 One-Parameter Toy Model with Finite $Ri_c$

Under the explicit assumption of equal profile parameters ( $\beta_m = \beta_h = \beta \implies Pr_t = 1$ ), the transformation reduces to  $Ri_g = \frac{\zeta}{1 + \beta\zeta}$ . Inverting directly gives:

$$\zeta(Ri_g) = \frac{Ri_g}{1 - \beta Ri_g} \implies \phi_m(Ri_g) = \frac{1}{1 - \beta Ri_g} \quad (28)$$

Defining the critical Richardson number  $Ri_c \equiv 1/\beta$ , the closure functions reduce to:

$$f_m(Ri_g) = f_h(Ri_g) = \left(1 - \frac{Ri_g}{Ri_c}\right)^2 = \left(\frac{Ri_c - Ri_g}{Ri_c}\right)^2 \quad (29)$$

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## 6 Surface Layer Roughness ( $z_0$ ) Integration

In numerical mesoscale and boundary-layer models (e.g., Louis 1979; McNider & Pielke 1981; England & McNider 1995), the finite vertical grid spacing requires calculating fluxes across a discrete layer from surface roughness height  $z_0$  to first model level  $z_1$ . The bulk Richardson number across this layer is defined as:

$$Ri_b = \frac{g}{\theta_0} \frac{[\theta(z_1) - \theta(z_0)](z_1 - z_0)}{[u(z_1) - u(z_0)]^2} \quad (30)$$

Integrating continuous MOST profile equations  $\frac{\partial u}{\partial z} = \frac{u_*}{\kappa z} \phi_m$  and  $\frac{\partial \theta}{\partial z} = \frac{\theta_*}{\kappa z} \phi_h$  from  $z_0$  to  $z_1$  yields the integrated profiles:

$$u(z_1) = \frac{u_*}{\kappa} \left[ \ln\left(\frac{z_1}{z_0}\right) - \Psi_m\left(\frac{z_1}{L}\right) \right], \quad \theta(z_1) - \theta(z_0) = \frac{\theta_*}{\kappa} \left[ \ln\left(\frac{z_1}{z_0}\right) - \Psi_h\left(\frac{z_1}{L}\right) \right] \quad (31)$$

Substituting these integrated profiles into (30) (assuming  $z_1 \gg z_0$  such that  $z_1 - z_0 \approx z_1$ ) expresses  $Ri_b$  in terms of integrated stability functions  $\Psi_m$  and  $\Psi_h$ :

$$Ri_b = \left(\frac{z_1}{L}\right) \frac{\ln(z_1/z_0) - \Psi_h(z_1/L)}{[\ln(z_1/z_0) - \Psi_m(z_1/L)]^2} \quad (32)$$

To ensure exact mathematical equivalence between local point gradient functions  $Ri_g = \zeta \frac{\phi_h}{\phi_m^2}$  and discrete layer computations  $Ri_b$ , an **effective depth**  $z_{\text{eff}}$  is defined such that  $Ri_g(z_{\text{eff}}/L) = Ri_b$ :

$$z_{\text{eff}} = z_1 \left[ \frac{\ln(z_1/z_0) - \Psi_m(z_1/L)}{\ln(z_1/z_0) - \Psi_h(z_1/L)} \right] \quad (33)$$

This formulation allows local  $Ri$ -based stability closures  $f_m(Ri)$  and  $f_h(Ri)$  to be evaluated directly using  $Ri_b$  at height  $z_{\text{eff}}$  without iterative root-finding.

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## A Complete Intermediate Derivation of the Unstable Cubic Equation

This appendix provides every intermediate algebraic step for deriving the unstable master cubic equation (21).

**Step 1: Squaring the Corrected Transformation Identity.** Starting from the corrected relation (7):

$$\zeta = Ri \frac{f_h(Ri)}{f_m^{3/2}(Ri)} \implies \zeta^2 = Ri^2 \frac{f_h^2(Ri)}{f_m^3(Ri)} \quad (A1)$$

**Step 2: Expressing  $\zeta$  in terms of  $f_m$ .** For Businger–Dyer momentum profile  $\phi_m = (1 - \beta_m \zeta)^{-1/4}$ , equation (18) gives:

$$f_m = (1 - \beta_m \zeta)^{1/2} \implies f_m^2 = 1 - \beta_m \zeta \implies \zeta = \frac{1 - f_m^2}{\beta_m} \quad (A2)$$

**Step 3: Expressing  $f_h^2$  in terms of  $f_m$ .** For heat profile  $\phi_h = (1 - \beta_h \zeta)^{-1/2}$ , equation (19) gives:

$$f_h = (1 - \beta_h \zeta)^{1/2} (1 - \beta_m \zeta)^{1/4} = (1 - \beta_h \zeta)^{1/2} f_m^{1/2} \quad (A3)$$

Substituting  $\zeta = \frac{1 - f_m^2}{\beta_m}$  into  $(1 - \beta_h \zeta)$ :

$$1 - \beta_h \zeta = 1 - \frac{\beta_h}{\beta_m} (1 - f_m^2) = \frac{\beta_m - \beta_h (1 - f_m^2)}{\beta_m} \quad (A4)$$

Squaring  $f_h$ :

$$f_h^2 = f_m \left[ \frac{\beta_m - \beta_h + \beta_h f_m^2}{\beta_m} \right] \quad (A5)$$

**Step 4: Substituting into Transformation Identity.** Substituting  $\zeta^2$  from (A2) and  $f_h^2$  from (A5) into (A1):

$$\left( \frac{1 - f_m^2}{\beta_m} \right)^2 = Ri^2 \frac{f_m \left( \frac{\beta_m - \beta_h + \beta_h f_m^2}{\beta_m} \right)}{f_m^3} \quad (A6)$$

Simplifying powers of  $f_m$  and  $\beta_m$ :

$$\frac{(1 - f_m^2)^2}{\beta_m^2} = \frac{Ri^2}{\beta_m f_m^2} [(\beta_m - \beta_h) + \beta_h f_m^2] \quad (A7)$$

**Step 5: Change of Variable  $x \equiv f_m^2$ .** Setting  $x = f_m^2$ :

$$\frac{(1 - x)^2}{\beta_m^2} = \frac{Ri^2}{\beta_m x} [(\beta_m - \beta_h) + \beta_h x] \quad (A8)$$

Multiplying both sides by  $\beta_m^2 x$ :

$$x(1 - x)^2 = Ri^2 \beta_m [(\beta_m - \beta_h) + \beta_h x] \quad (A9)$$

Expanding the left-hand side:

$$x(1 - 2x + x^2) = x^3 - 2x^2 + x \quad (A10)$$

Expanding the right-hand side and grouping terms:

$$x^3 - 2x^2 + x = Ri^2 \beta_m (\beta_m - \beta_h) + Ri^2 \beta_m \beta_h x \quad (A11)$$

Bringing all terms to the left-hand side yields equation (21):

$$x^3 - 2x^2 + (1 - Ri^2 \beta_m \beta_h)x - Ri^2 \beta_m (\beta_m - \beta_h) = 0 \quad (A12)$$

This completes the formal proof.

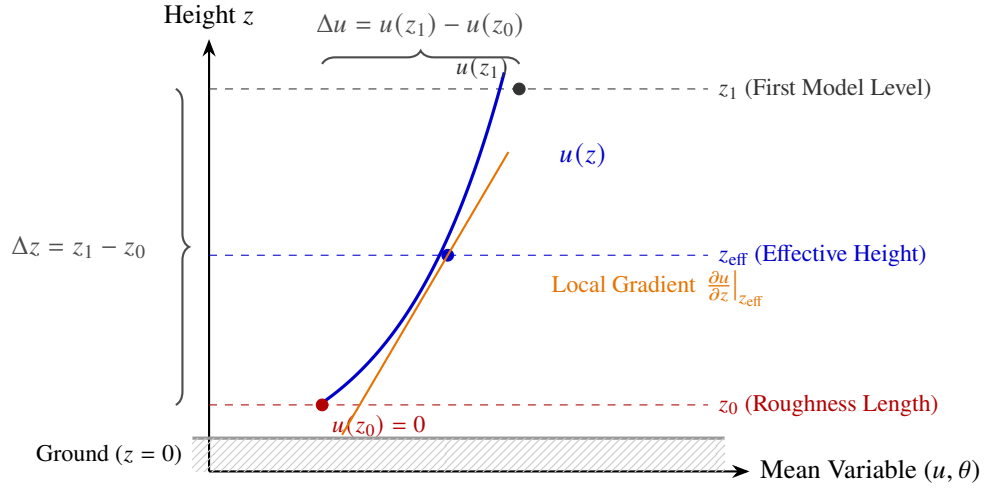


Figure 1: Schematic of discrete surface layer discretization illustrating the bulk layer differences over  $\Delta z$  and the effective height  $z_{\text{eff}}$  matching the local point gradient  $Ri_g$  to the bulk layer  $Ri_b$ .

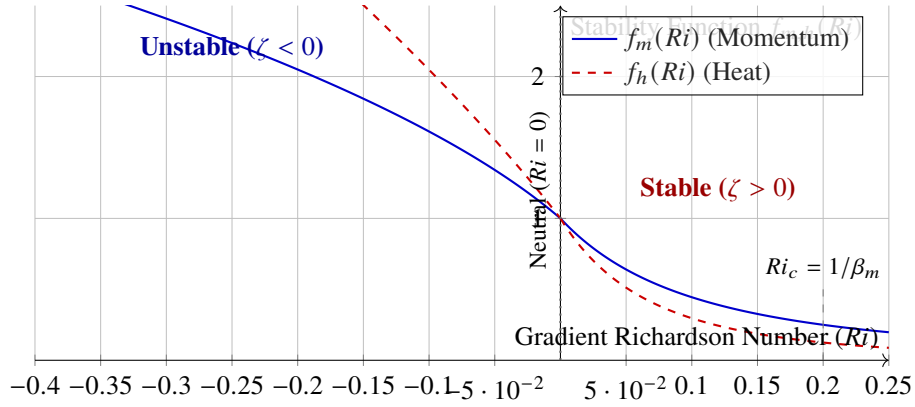


Figure 2: Behavior of momentum  $f_m(Ri)$  and heat  $f_h(Ri)$  stability functions across unstable ( $Ri < 0$ ) and stable ( $Ri > 0$ ) regimes showing the asymptotic decay toward critical Richardson number  $Ri_c$ .

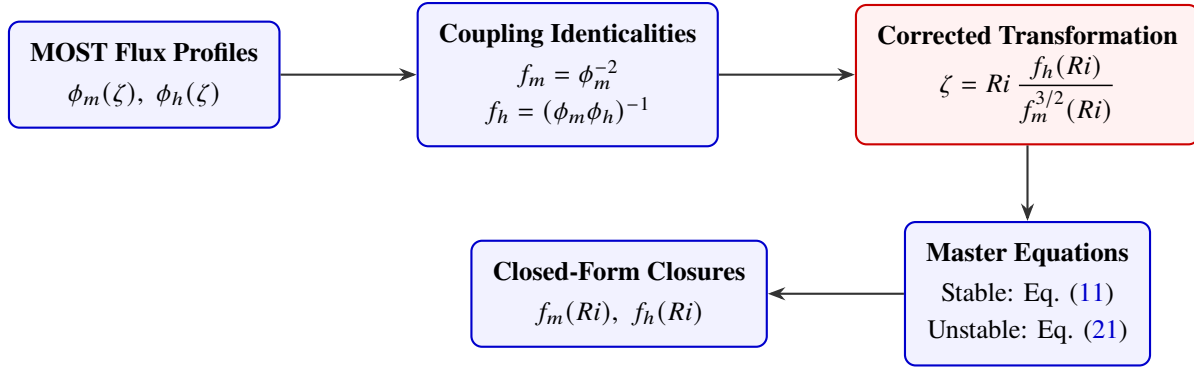


Figure 3: Mathematical workflow for transforming Monin–Obukhov Similarity Theory profile functions  $\phi_{m,h}(\zeta)$  into analytical gradient Richardson number stability functions  $f_{m,h}(Ri)$ .

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